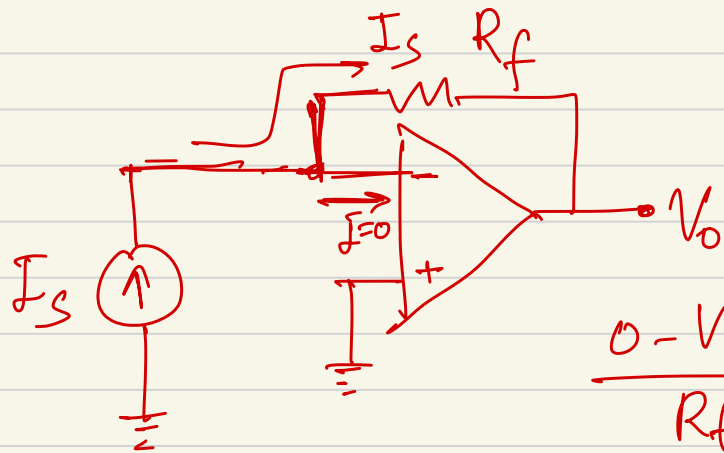


Tutorial-7:-

Q1. You are building a barcode scanner. The detector is a photodiode that produces a tiny current proportional to light intensity, ranging from $0 \mu\text{A}$ (dark) to $50 \mu\text{A}$ (bright light). The microcontroller's analog-to-digital converter (ADC) requires a voltage input of 0 V to 5 V .

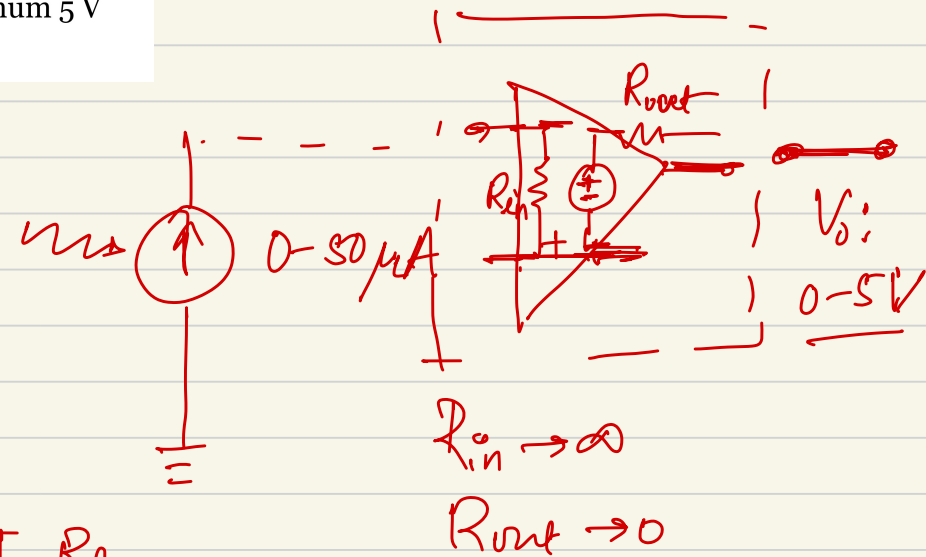
Design a Current-to-Voltage converter using Op-AMP to achieve the maximum 5 V output when the sensor outputs $50 \mu\text{A}$.



$$\frac{0 - V_o}{R_f} = I_s$$

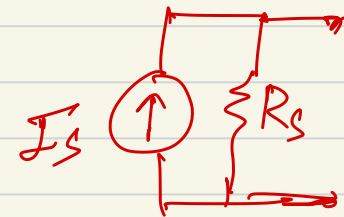
$$V_o = I_s R_f$$

$$R_f = \frac{5\text{V}}{50\mu\text{A}} = 100\text{k}\Omega$$

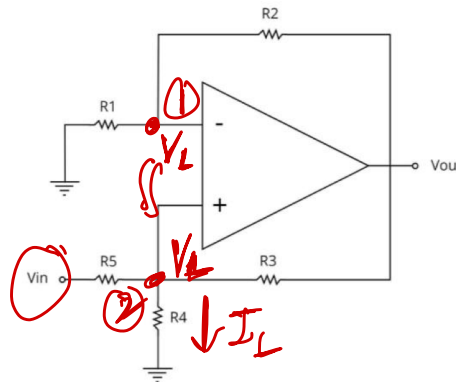


Extension

If photodiode has an internal resistance :-
Redesign



Q2. The microcontroller output (V_{in}) ranging from 0 V to +5 V is used a control signal to directly drive an industrial proportional solenoid valve to regulate fluid flow. The valve requires a drive current (I_L) of 0 to 200 mA to operate fully. You implement the shown Voltage-to-Current converter circuit, where R_4 represents the resistance of the valve and V_L denotes the voltage across the valve (at the non-inverting terminal).



- Derive the equation for the load current (I_L) in terms of V_{in} , V_L , and the resistors (R_1 , R_2 , R_3 , and R_5).
- For this circuit to act as a true current source, the current delivered to the valve (I_L) must not be affected by the valve's resistance or the voltage across it (V_L). Based on your equation from A, what specific mathematical ratio must exist between the resistors to cancel out the V_L term?
- Choosing $R_1 = 10 \text{ k}\Omega$ and $R_2 = 20 \text{ k}\Omega$, calculate the exact values required for R_3 and R_5 to ensure the circuit is perfectly balanced and delivers exactly 200 mA to the valve when V_{in} is +5 V.

$$V_{in} : 0 - 5 \text{ V}$$

$$I_L : 0 - 200 \text{ mA}$$

node ① KCL:

$$\frac{V_L - 0}{R_1} + \frac{V_L - V_{out}}{R_2} = 0$$

$$V_L \left[\frac{1}{R_1} + \frac{1}{R_2} \right] = -\frac{V_{out}}{R_2}$$

$$\Rightarrow V_{out} = V_L \left[1 + \frac{R_2}{R_1} \right]$$

KCL at node 2:-

$$\frac{V_L - V_{in}}{R_5} + I_L + \frac{V_L - V_{out}}{R_3} = 0$$

$$I_L = +\frac{V_{in}}{R_5} - V_L \left[\frac{1}{R_3} + \frac{1}{R_5} \right] + \frac{V_{out}}{R_3}$$

$$= \frac{V_{in}}{R_5} - V_L \frac{R_3 + R_5}{R_3 R_5} + V_L \frac{(R_1 + R_2)}{R_1 R_3}$$

$$I_L = \frac{V_{in}}{R_5} + V_L \left[\frac{R_1 + R_2}{R_1 R_3} - \frac{R_3 + R_5}{R_3 R_5} \right]$$

B) for I_L to be independent of V_L :-

$$\frac{R_1 + R_2}{R_1 R_3} = \frac{R_3 + R_5}{R_3 R_5} \Rightarrow I_L = \frac{V_{in}}{R_5}$$

$$1 + \frac{R_2}{R_1} = 1 + \frac{R_3}{R_5} \Rightarrow \boxed{\frac{R_2}{R_1} = \frac{R_3}{R_5}}$$

$$\text{or } \frac{R_1}{R_2} = \frac{R_5}{R_3}$$

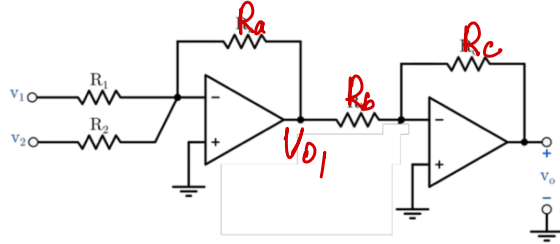
C) $I_L = 200 \text{ mA} @ V_{in} = 5 \text{ V} \Rightarrow R_5 = \frac{V_{in}}{I_L} = \frac{5}{0.2} = 25 \Omega$

$R_1 = 10 \text{ k}\Omega, R_2 = 20 \text{ k}\Omega$

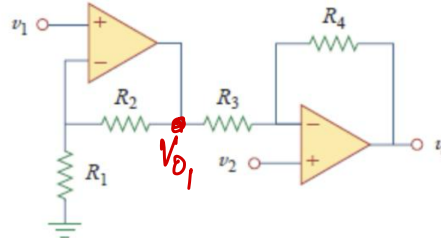
$$\frac{R_2}{R_1} = \frac{R_3}{R_5} \Rightarrow R_3 = R_5 \times \frac{R_2}{R_1} = 2 \times 25 = \underline{\underline{50 \Omega}}$$

Q3: You are designing an analog audio mixer to combine a vocal microphone (v_1) and a backing instrument track (v_2). v_1 needs to be amplified by a gain of 5 while v_2 is already loud and needs its volume reduced by a gain of 0.5. Two junior engineers pitch different 2-Op-Amp circuit topologies to achieve this:

Design A



Design B



- Derive the final output voltage equation v_o for both designs terms of v_1 and v_2 , and the resistors.
- A DJ needs to adjust the gain of the backing track (v_2) without changing the volume of the vocals (v_1). Can you compare both designs on the feasibility of adjusting the gain of v_2 independently?
- A good audio mixer must combine signals additively so they don't cancel each other out. Which design fails to mix the audio additively, and why?
- Find the value of resistors for the chosen design to achieve required functionality.

For Design A

$$V_{O1} = -\frac{R_a}{R_1} U_1 - \frac{R_a}{R_2} U_2$$

$$V_o = -\frac{R_c}{R_b} \cdot V_{O1}$$

$$V_o = +\frac{R_c}{R_b} \left(\frac{R_a}{R_1} U_1 + \frac{R_a}{R_2} U_2 \right)$$

for design B

$$V_{O1} = \left(1 + \frac{R_2}{R_1} \right) U_1$$

for second opamp :-

super position :-

$$V_o = V_o' (U_2 \text{ is active}) + V_o'' (V_{O1} \text{ is active})$$

$$V_o = \left(1 + \frac{R_4}{R_3} \right) U_2 - \frac{R_4}{R_3} \cdot V_{O1}$$

$$\Rightarrow \boxed{V_o = \left(1 + \frac{R_4}{R_3}\right) V_2 - \frac{R_4}{R_3} \left(1 + \frac{R_2}{R_1}\right) V_1}$$

B) for design B, to adjust gain of V_2 , R_4/R_3 needs to be modified which will also modify gain of V_1 .

for design A, R_2 can be adjusted to tune the gain of V_2 without modifying gain of V_1 .

C) for V_1 & V_2 to additive, we choose design A.

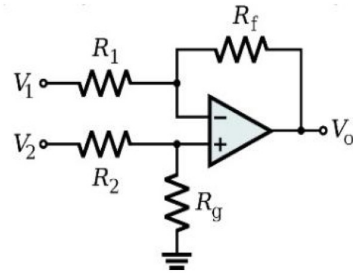
→ Design B is subtracting V_2 & V_1 rather than adding.

d) $V_o = \frac{R_c}{R_b} \left(\frac{R_a}{R_1} V_1 + \frac{R_a}{R_2} V_2 \right) \rightarrow \text{Design A.}$

$$\frac{R_c R_a}{R_1 R_b} = 5, \quad \frac{R_c}{R_b} \cdot \frac{R_a}{R_2} = 0.5, \quad \frac{R_2}{R_1} = 10, \quad \text{Chosen } R_1 = 1 \text{ k}\Omega, \quad R_2 = 10 \text{ k}\Omega, \quad \frac{R_c R_a}{R_b} = 5$$

→ Do not choose very low values for R_1, R_2 & R_b as it affects the input impedance.

Q4: The circuit shown is a difference amplifier, widely used to read signals from sensors. In this problem, we will understand its functionality, design it to reject electrical noise, and see what happens when we connect a real-world sensor to it.



- Find the output voltage (V_{out}) in terms of V_1 , V_2 , and the four resistors [Hint: May use Superposition Theorem].
- You connect a sensor across the two inputs. The sensor outputs a tiny differential signal (V_d) but both wires also pick up the exact same amount of background electrical noise from the room (V_{noise}). This means the inputs are: $V_1 = V_{noise}$ and $V_2 = V_{noise} + V_d$. To perfectly reject the noise so that V_{noise} disappears from your final output, what specific ratio must your resistors satisfy?
- Real sensors have an internal source resistance (R_s) that forms a voltage divider with the amplifier's total input resistance ($R_{in} = R_1 + R_2$). If you connect a sensor where $R_s = R_1 + R_2$, what fraction of the original differential signal (V_d) actually survives to reach the amplifier?

$$V_{out} = V_{o1} + V_{o2}$$

V_{o1} (V_1 is active) } Superposition
 V_{o2} (V_2 is active)

$$V_{o1} = -\frac{R_f}{R_1} V_1$$

$$V_{o2} = \frac{R_g}{R_g + R_2} \cdot V_2 \cdot \left(1 + \frac{R_f}{R_1}\right)$$

$$= \frac{R_g}{R_1} \left(\frac{R_1 + R_f}{R_g + R_2}\right) \cdot V_2$$

$$\Rightarrow V_{out} = V_{o1} + V_{o2}$$

$$V_{out} = -\frac{R_f}{R_1} V_1 + \frac{R_g}{R_1} \left(\frac{R_1 + R_f}{R_g + R_2}\right) V_2$$

B) $V_1 = V_{noise}$, $V_2 = V_{noise} + V_d$, to cancel V_{noise} in the V_o expression:-

$$\frac{R_g}{R_1} \left(\frac{R_1 + R_f}{R_g + R_2}\right) = \frac{R_f}{R_1}$$

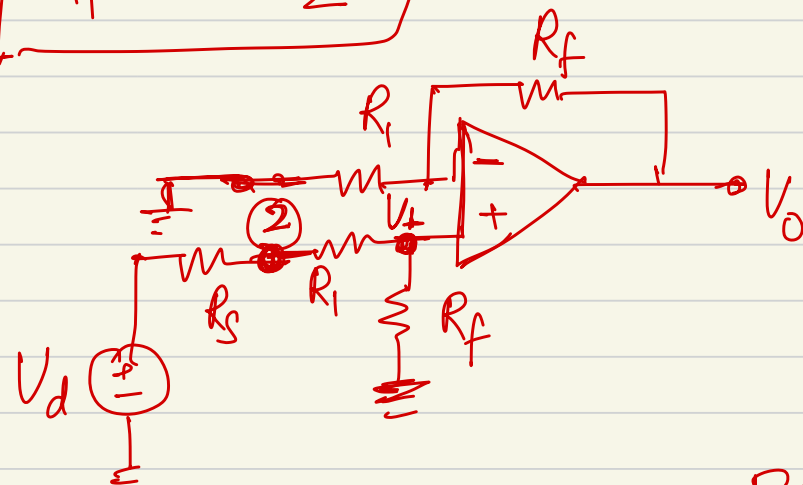
$$1 + \frac{R_1}{R_f} = 1 + \frac{R_2}{R_g} \Rightarrow \frac{R_1}{R_f} = \frac{R_2}{R_g}$$

or $\boxed{\frac{R_f}{R_1} = \frac{R_g}{R_2}}$

Case if $R_2 = R_1$, $R_g = R_f$

c)

$$V_o = \left(1 + \frac{R_f}{R_1}\right) \cdot \frac{R_f}{R_f + R_s + R_1} \cdot V_d$$

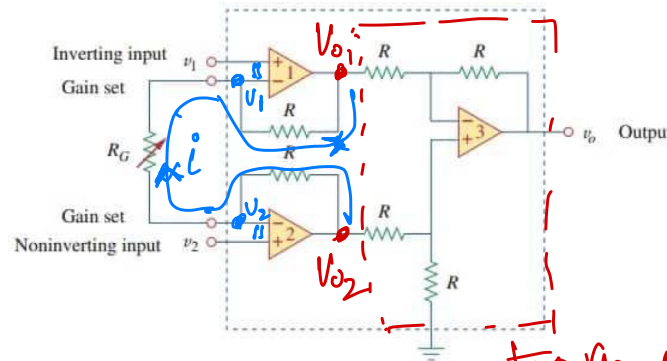


$$V_+ = \frac{R_f}{R_f + R_s + R_1} V_d \quad , \text{ if } R_s \text{ was not present, } V_+ = \frac{R_f}{R_f + R_1} V_d$$

$$R_{\text{input from node 2}} = R_1 + R_f$$

$$\text{if } R_s = R_f + R_1 \Rightarrow V_+ = \frac{R_f}{2(R_f + R_1)} V_d \quad \left(\text{it is half of the signal when } R_s = 0 \right)$$

Q5: Measuring a human heartbeat (ECG) requires reading a tiny, microvolt-level differential signal from electrodes placed on the skin. You must not draw any current from the patient's body, as doing so will physically alter the voltage reading (the loading effect we saw in Q4). Because a standard 1-Op-Amp difference amplifier has a low input impedance, you select a classic 3-Op-Amp Instrumentation Amplifier as shown here.



→ Differential Amplifier

- Explain briefly why the two input buffer Op-Amps solve the input impedance problem.
- Biomedical signals are very small, so your ECG circuit needs a total differential gain of 51. Calculate the value of the external gain-setting resistor (R_G) required to achieve this.

A. As V_1 & V_2 are directly applied to the non-inverting terminal of Op-Amp. The input resistance can be assumed to be ∞ compared to Q4 (where it was $R_1 + R_1$).

B. $A_d = 51$, $\frac{U_0}{V_1 - V_2} = 51$

from Q4: $U_0 = -\frac{R_f}{R_1} U_{01} + \frac{R_g}{R_1} \left(\frac{R_1 + R_f}{R_2 + R_g} \right) U_{02}$

$R_1 = R_2 = R_f = R_g = R \Rightarrow U_0 = -U_{01} + U_{02} = U_{02} - U_{01}$

$i = \frac{U_2 - U_1}{R_g}$, path from U_{02} to $U_{01} \Rightarrow U_{02} - iR - iR_g - iR - U_{01} = 0$

$$\Rightarrow U_{02} - U_{01} = 2iR + iR_g = (2R + R_g) \frac{(U_2 - U_1)}{R_g}$$

$$\Rightarrow \frac{U_{02} - U_{01}}{U_2 - U_1} = \left(1 + \frac{2R}{R_g}\right)$$

$$\frac{U_0}{U_2 - U_1} = \frac{U_{02} - U_{01}}{U_2 - U_1} = \left(1 + \frac{2R}{R_g}\right) = 51 \quad (\text{required})$$

$$R_g = \frac{2R}{50}, \quad \text{if } R = 10 \text{ k}\Omega, \Rightarrow R_g = \frac{20 \text{ k}}{50} = 0.4 \text{ k}\Omega$$